Home@ix FAIR: A Forward Indicator of UK Housing Affordability Regimes Using Credit–Price Decoupling and Market Depth

Home@ix (Working Paper / Draft)

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Abstract

UK housing affordability is usually discussed as a price level or a price-to-income ratio. That framing is necessary but incomplete because UK housing markets often do not clear primarily through price adjustment. In stressed regimes, markets clear through quantities and composition: turnover collapses, chains fail, mortgage-dependent households are rationed out, and observed prices can appear sticky or misleading. This paper introduces Home@ix FAIR: a quarterly indicator designed as a 2–3 year forward regime signal for affordability understood as access and allocation, not only price burden. FAIR is intentionally reduced-form and reproducible. It integrates a credit–price wedge (prices outrunning mortgage-stock growth), a market-depth term (turnover dynamics) capturing thin-market clearing, and an optional composition term (new-build share dynamics). To avoid defining "normal" using crisis regimes and structural breaks, FAIR is normalised to pooled baseline windows excluding the Global Financial Crisis aftermath and Covid-era disruptions. FAIR is paired with a direction-of-flow statistic and can be evaluated via event-based backtesting (drawdown-defined crash starts). Together, the index and backtests form an operational toolkit for monitoring affordability regime risk.

1 Executive Summary

UK housing affordability is typically treated as a static burden ratio. Yet housing is not a frictionless market. When credit tightens or uncertainty rises, the UK housing market frequently clears via reduced participation rather than orderly repricing: transactions fall, turnover collapses, and access deteriorates even when price ratios appear stable.

This paper introduces **Home@ix FAIR**, a reproducible quarterly indicator designed to act as a **2–3 year forward regime signal** for affordability understood as **access and allocation**. FAIR integrates:

- a **credit–price wedge** capturing decoupling between house-price growth and mortgage-stock growth;
- a **market-depth / turnover term** capturing thin-market (quantity-clearing) stress; and
- an **optional composition term** based on new-build share dynamics.

To avoid contaminating baseline moments with crisis regimes and structural breaks, FAIR is normalised using pooled baseline windows:

- **Baseline windows:** 2003Q1–2007Q4 and 2013Q1–2019Q4
- **Excluded:** 2008–2012 (GFC and aftermath) and 2020+ (Covid and subsequent regime breaks)

FAIR is accompanied by a direction-of-flow measure and can be assessed as an early-warning tool using objective event definitions (e.g., crash starts defined by drawdowns from local peaks).

2 Introduction: Affordability as Access, Allocation, and Market Clearing

Standard affordability measures (e.g., price-to-income ratios) treat affordability as a price-only object. But UK housing often fails to clear under stress through price adjustment. Instead, the system shifts toward thin-market clearing:

- transactions decline sharply;
- the transacting population becomes more selected (composition shifts);
- access deteriorates even if headline price ratios appear stable.

This paper therefore treats affordability as a **regime outcome** shaped by credit creation, market depth, and institutional constraints, and introduces an indicator designed to detect when the system is moving into a regime where affordability is likely to deteriorate over the next 2–3 years.

3 From Descriptive Diagnosis to a Forward-Looking Tool

Home@ix begins from an integrated diagnosis: mortgage-credit dominance in bank balance sheets, endogenous money creation via bank lending, and housing's role as both a necessity and collateral asset. Moving from diagnosis to a forward-looking tool requires operationalising how stressed regimes appear in observable data.

3.1 Why "forward-looking" is operationally meaningful here

A forward-looking affordability signal must capture two patterns that commonly precede later deterioration:

- **Credit–price decoupling:** price growth outruns mortgage-stock growth.
- **Thin-market clearing:** market depth collapses (turnover declines), indicating rationing and throughput failure.

FAIR integrates these mechanisms into a single auditable index with an explicit baseline definition.

4 Data and Reproducible Construction

FAIR is computed from quarterly series produced by the Home@ix feature set.

4.1 Inputs (quarterly)

Key variables:

- House prices (level): `avg_house_price_gbp`
- Mortgage stock (level, £m): `mb_total_gbp_m`
- Turnover (market depth): `turnover_pct_q` (transactions scaled by dwelling stock)
- Optional composition proxy: `newbuild_share_of_transactions`
- Time index: `period` (quarter), and `geo` for region selection (e.g., EW)

4.2 Derived series

Define year-on-year (YoY) growth rates:

$$g_t^P = YoY(P_t)$$

$$g_t^{MB} = YoY(MB_t)$$

$$g_t^{TO} = YoY(TO_t)$$

Define the credit–price wedge:

$$W_t = g_t^P - g_t^{MB}$$

Optional new-build share change (YoY):

$$\Delta NB_t = NB_t - NB_{t-4}$$

5 Baseline Normalisation (Core Proviso)

To avoid defining "normal" using crisis regimes and structural breaks, z-scores are computed from pooled baseline windows:

- 2003Q1–2007Q4
- 2013Q1–2019Q4

Exclusions:

- 2008–2012 (Global Financial Crisis and aftermath)
- 2020+ (Covid and subsequent regime breaks)

For any series X_t, define:

$$z(X_t) = \frac{X_t - \mu_X}{\sigma_X}$$

where

$$\mu_X$$

and

$$\sigma_X$$

are computed using baseline quarters only.

6 The Home@ix FAIR Indicator

6.1 FAIR definition

Default (three-component) specification:

$$FAIR_t = 100 \cdot \Big(0.55 \cdot z(W_t) - 0.35 \cdot z(g_t^{TO}) + 0.10 \cdot z(\Delta NB_t)\Big)$$

If the new-build term is not available, the final term is omitted:

$$FAIR_t = 100 \cdot \Big(0.55 \cdot z(W_t) - 0.35 \cdot z(g_t^{TO})\Big)$$

6.2 Direction-of-flow

Define the direction-of-flow statistic:

$$\Delta FAIR_t = FAIR_t - FAIR_{t-1}$$

This is used for monitoring whether affordability pressure is accelerating or easing.

6.3 Interpretation bands (practical)

A simple interpretive scheme:

- $$FAIR \geq 50$$

  : strong deterioration regime risk

- $$20 \leq FAIR < 50$$

  : mild deterioration bias

- $$-20 \leq FAIR < 20$$

  : neutral/noisy

- $$-50 \leq FAIR < -20$$

  : mild improvement bias

- $$FAIR < -50$$

  : strong improvement regime (often stress-driven)

7 Using FAIR for Forward-Looking Evaluation (Event Backtesting)

FAIR can be evaluated using objective events and simple warning rules.

7.1 Crash-start events (rule-based)

A "crash start" may be defined as a local price peak followed by a drawdown of at least

$$X\%$$

within

$$Y$$

quarters, with a minimum separation (cooldown) of

$$C$$

quarters to avoid double-counting clustered peaks.

7.2 Illustrative warning rules

Examples:

- A: sustained level stress:

  $$FAIR > 20$$

  for 2 quarters

- B: sustained acceleration:

  $$\Delta FAIR > 5$$

  for 2 quarters

- C: broad stress-worsening:

  $$FAIR > 0$$

  and

  $$\Delta FAIR \geq 0$$

7.3 Metrics

- Lead time (quarters) from the most recent signal to each crash start.
- False-positive proxy: share of signal quarters not followed by any crash start within a forward window (e.g., 8 quarters).

8 Outputs and Visualisations (Reproducible)

The FAIR build produces:

- a FAIR level figure (headline chart),
- a component contributions figure (mechanism chart),
- an animated direction-of-flow GIF emphasising

$$\Delta FAIR_t$$

,

- and a tidy CSV export of intermediate series for auditability.

8.1 Figure placeholders

Replace the filenames with the paths to the generated assets.

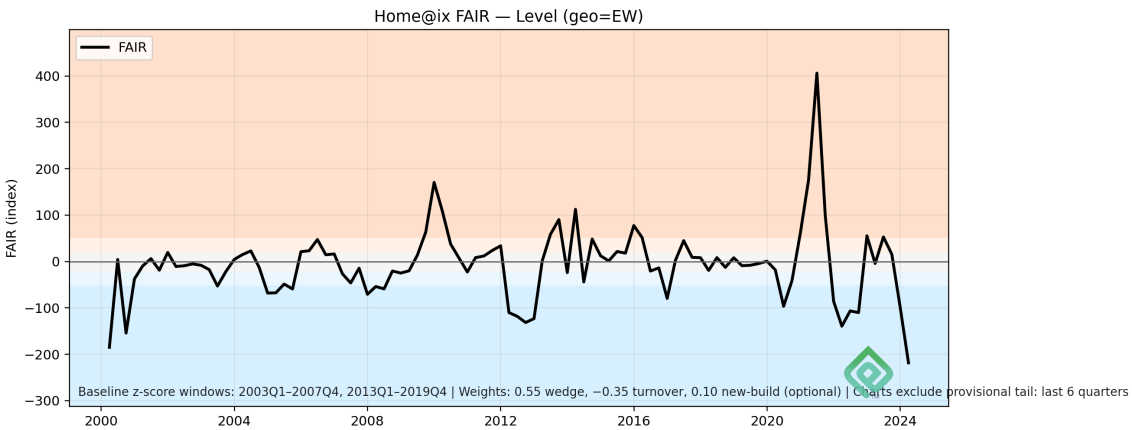

Figure 1: Home@ix FAIR (EW): Level. Baseline windows: 2003Q1–2007Q4 and 2013Q1–2019Q4; excludes 2008–2012 and 2020+.

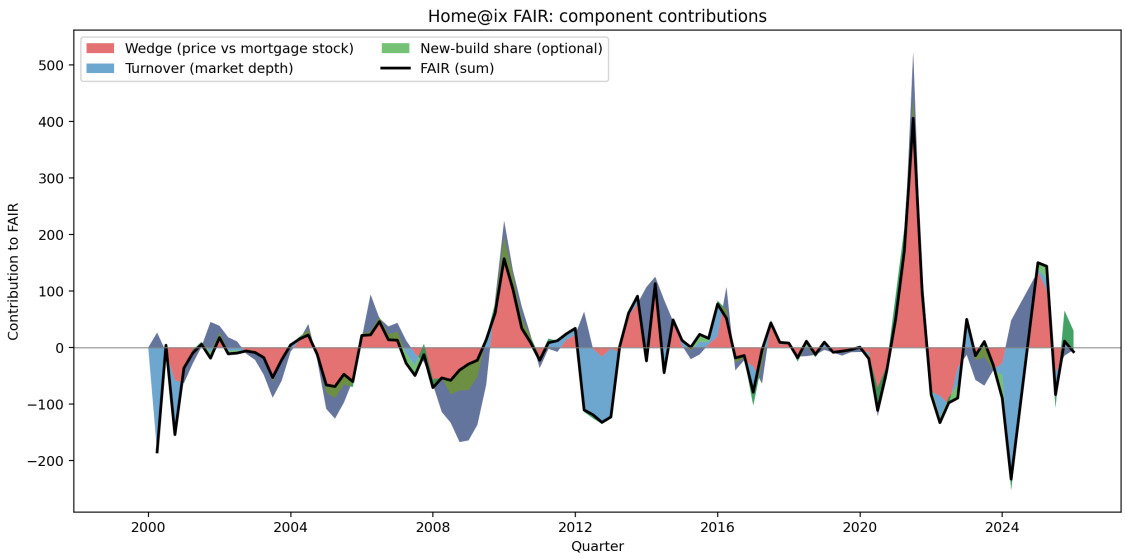

Figure 2: Home@ix FAIR (EW): Component contributions. FAIR is the sum of wedge, turnover, and optional new-build terms.

9 Limitations and Extensions

FAIR is a reduced-form indicator derived from aggregate market series. It does not directly encode distributional constraints (income heterogeneity, cash-buyer share, arrears microdata) or direct developer pacing series (e.g., sales per outlet per week) unless explicitly added. These are natural extensions; the baseline FAIR specification is intentionally reproducible and captures two dominant UK mechanisms: credit–price decoupling and thin-market clearing.

10 Conclusion

UK housing affordability is not only a price level problem; it is also an allocation and throughput problem. Home@ix FAIR provides a transparent, reproducible way to detect when the system is shifting into a regime where affordability is likely to deteriorate over a 2–3 year horizon via credit–price decoupling and/or collapsing market depth. Combined with direction-of-flow and event backtesting, FAIR is positioned as an operational forward-looking tool: a single "Number" grounded in auditable data and interpretable mechanisms.

Reproducibility Note

The figures referenced in Section 8 are produced by the FAIR build scripts in the Home@ix workflow. The output directory should contain:

- `fair_quarterly_audit.csv`
- `fig_fair_level.png`
- `fig_fair_contributions.png`

- `anim_fair_direction_of_flow.gif`